

Training & Build Schedule — SIH to LaunchpadX

Goal: Train teammates on agentic AI + actually build agents together

Today: Aug 10, 2026 (Sunday)

SIH: Aug 18-19 (Monday-Tuesday)

LaunchpadX: Aug 22-23 (Friday-Saturday) @ SNIST Hyderabad

Theme: Agentic AI / GenAI / Agent Building

Overview (13 Days)

Aug 10-12	PHASE 1: LEARN (Theory + Concepts)
Aug 13-15	PHASE 2: BUILD (Hands-on Agent Building)
Aug 16-17	PHASE 3: PREP SIH (Polish + Demo Practice)
Aug 18-19	■■■ SIH HACKATHON ■■■
Aug 20-21	PHASE 4: LEVEL UP (Advanced + LaunchpadX Prep)
Aug 22-23	■■■ LAUNCHPADX HACKATHON ■■■

PHASE 1: LEARN — Theory & Concepts

Day 1 — Sun Aug 10: "What is Agentic AI?"

Morning (2-3 hrs) — Theory Session

Topic	Resource	Duration
What is an AI agent vs chatbot	Agentic AI - Master Guide	30 min
6 types of agents (Russell & Norvig)	Types of AI Agents	30 min
7 design patterns (ReAct, Reflection, Planning...)	Agent Design Patterns	45 min
Multi-agent architectures (Pipeline, Swarm, DAG)	Multi-Agent Architectures	30 min
Q&A / Discussion	—	15 min

Afternoon (2 hrs) — Show & Tell

Activity	Details
Demo our AI Software Company	Run the full pipeline live — problem → deliverables
Walk through the code	Show how 6 agents are orchestrated in <code>engine.py</code>
Map concepts to our project	Use Our AI Software Company as the guide
Assign reading	Each teammate reads 2 knowledge pages before tomorrow

Homework:

- [] Everyone reads Agent Frameworks Comparison and Agent Protocols - MCP and A2A
- [] Each person picks a framework they want to try (n8n, Claude SDK, CrewAI, or LangGraph)

Day 2 — Mon Aug 11: "How Agents Connect to the World"

Morning (2-3 hrs) — Protocols & Tools

Topic	Resource	Duration
MCP — "USB-C for AI"	Agent Protocols - MCP and A2A	30 min
A2A — agent-to-agent communication	Agent Protocols - MCP and A2A	20 min
RAG — how agents access knowledge	RAG - Retrieval Augmented Generation	45 min
Voice & Call agents	Voice Agents + Call Agents	30 min
The market — who's building what	AI Agent Market - Competitors	20 min

Afternoon (2-3 hrs) — Hands-on: First Agent

Activity	Details
Install tools	Python 3.11+, Node.js, n8n (Docker), Claude CLI
Build "Hello Agent"	Simple ReAct agent using OpenAI/Claude API — ask question, use web search tool, answer
Everyone builds one	Each person writes a 30-line agent that uses at least 1 tool

Code template for first agent:

```
# hello_agent.py – Everyone's first agent
import openai

client = openai.OpenAI(base_url="https://openrouter.ai/api/v1", api_key=" ... ")

tools = [{
    "type": "function",
    "function": {
        "name": "search_web",
        "description": "Search the web for information",
        "parameters": {"type": "object", "properties": {"query": {"type": "string"}}}
    }
}]

response = client.chat.completions.create(
    model="openai/gpt-4o-mini",
    messages=[{"role": "user", "content": "What's the weather in Hyderabad today?"}],
    tools=tools
)
# Handle tool calls, execute, feed back – the agent loop!
```

Homework:

- [] Modify your agent to use 2 tools (search + calculator)
- [] Read How to Build AI Agents — pick which Level you want to go deeper on

Day 3 — Tue Aug 12: "Frameworks Deep Dive"

Morning (2-3 hrs) — Framework Workshop

Split into groups based on interest:

Group	Framework	What They Build	Guide
Group A	n8n	AI agent workflow on our n8n instance	How to Build AI Agents Level 2
Group B	Claude Agent SDK	Agent with subagents and MCP tools	How to Build AI Agents Level 3
Group C	CrewAI / LangGraph	Multi-agent crew with roles	How to Build AI Agents Level 3

Afternoon (2-3 hrs) — Build Sprint #1

Activity	Details
Each group builds a working agent	Must take input → reason → use tools → produce output
Cross-demo	Each group demos to the others (10 min each)
Discuss trade-offs	Which framework was easiest? Most powerful? Most flexible?

Deliverable: Each group has a working agent they can demo.

PHASE 2: BUILD — Hands-on Agent Building

Day 4 — Wed Aug 13: "Build a Multi-Agent System"

Full Day (5-6 hrs) — Team Build

Time	Activity
9:00-10:00	Design session — sketch a 3-agent pipeline on whiteboard
10:00-13:00	Build it — each person owns one agent
14:00-16:00	Integration — connect agents, test the pipeline
16:00-17:00	Debug and polish

Project idea: Build a mini version of our AI Software Company with 3 agents:

- 1. **Planner Agent** — takes a problem, creates a plan
- 2. **Researcher Agent** — searches the web, gathers info
- 3. **Writer Agent** — produces a report using plan + research

This teaches: agent handoffs, shared state, tool use, output parsing.

Day 5 — Thu Aug 14: "Add RAG + Advanced Features"

Morning (3 hrs) — RAG Workshop

Activity	Details
Set up a vector database	ChromaDB (local, easy) or Pinecone (cloud)
Embed documents	Chunk → embed → store the SIH problem statements
Build a RAG agent	Query the vector store, get grounded answers

Afternoon (3 hrs) — Advanced Patterns

Activity	Details
Add cross-review	Agent A reviews Agent B's output (our pattern)
Add approval gates	Human-in-the-loop: pause, show output, wait for approval
Add memory	Agent remembers context from previous interactions
Test with real problems	Use SIH problem statements as input

Day 6 — Fri Aug 15: "n8n Agent Workflows + Integration"

Morning (3 hrs) — n8n Deep Dive

Activity	Details
Connect to our n8n instance	srv1867770.hstgr.cloud
Build an AI Agent workflow	Webhook trigger → AI Agent → response
Add MCP tools	Connect MCP servers to n8n
Chain agents	Agent 1 output → Agent 2 input in n8n

Afternoon (3 hrs) — Build Something Demoable

Activity	Details
Each person builds one complete agent workflow in n8n	Must be demoable
Combine into a team showcase	"Here are 4 different agents we built in n8n"
Record a 2-min walkthrough	Screen record for reference

PHASE 3: PREP SIH

Day 7 — Sat Aug 16: "SIH Strategy & Polish"

Morning (3 hrs) — SIH Problem Analysis

Activity	Details
Review SIH problem statements	Pick top 3 candidates
Map each to our pipeline	Which ones work best with our system?
Design the solution approach	For each candidate, sketch the 6-agent pipeline
Prepare backup plans	What if the API is down? Demo mode ready?

Afternoon (3 hrs) — System Testing

Activity	Details
Full pipeline test run	Problem → all 6 agents → deliverables
Test demo/fallback mode	Cached responses work?
Fix any bugs	Last chance for code fixes
Prepare the dev environment	All API keys working, servers running

Day 8 — Sun Aug 17: "Demo Practice & Final Prep"

Morning (2 hrs) — Presentation Prep

Activity	Details
Write the pitch script	5-min pitch for judges
Assign speaking roles	Who presents what
Create the demo flow	Step-by-step what to show

Afternoon (3 hrs) — Rehearsal

Activity	Details
Full dry run #1	Time it, note issues
Fix issues	Adjust timing, fix demo glitches
Full dry run #2	Should be smooth
Backup plan rehearsal	Practice demo mode in case of failures
Pack for hackathon	Laptops charged, hotspot ready, extension cords

Focus: Execute. Build. Demo. Win.

Take notes on what judges ask — those insights help for LaunchpadX.

PHASE 4: LEVEL UP — LaunchpadX Prep

Day 9 — Wed Aug 20: "Post-SIH Debrief + LaunchpadX Strategy"

Morning (2 hrs) — Debrief

Activity	Details
What worked at SIH?	List wins
What didn't work?	List problems and fixes
Judge feedback	What did they ask? What impressed them?
Code fixes	Fix anything that broke during SIH

Afternoon (3 hrs) — LaunchpadX Pivoting

Activity	Details
Review LaunchpadX theme	"Agentic AI / GenAI / Agent Building"
Decide: enhance existing or build new?	Our AI Software Company IS an agent-building project
Plan new features for LaunchpadX	What can we add in 2 days?
Assign tasks	Each person owns a feature

LaunchpadX-specific features to consider:

- [] Live agent builder UI (drag-and-drop agent creation)
- [] Voice interface (Vapi integration for voice input)
- [] MCP server that exposes our pipeline as a tool
- [] Agent marketplace concept (browse pre-built agent templates)
- [] Real-time agent visualization (watch agents think)

Day 10 — Thu Aug 21: "Build for LaunchpadX"

Full Day (6-8 hrs) — Build Sprint

Time	Activity
9:00-10:00	Finalize feature list for LaunchpadX
10:00-13:00	Build sprint — everyone codes
14:00-17:00	Integration + testing
17:00-18:00	Demo rehearsal (LaunchpadX version)
18:00-19:00	Pack, prep, final checks

Pitch angle for LaunchpadX:

"We built an AI Software Company — a team of 6 AI agents that work as a startup. But more than that, we've mastered every approach to building agents: no-code with n8n, SDK with Claude Agent SDK, and custom orchestration. We didn't just build one agent — we built an agent factory."

LAUNCHPADX HACKATHON — Aug 22-23 (Fri-Sat) @ SNIST

Theme: Agentic AI / GenAI / Agent Building

Pitch: We ARE agent builders. Our project IS the proof.

Daily Checklist Template

Use this for each training day:

- ☐ Morning standup (5 min) — what are we learning/building today?
- ☐ Theory/reading done
- ☐ Hands-on exercise completed
- ☐ Each person can explain what they built
- ☐ Code committed / workflow saved
- ☐ Evening review — what clicked, what's confusing?

Tools Everyone Needs Installed

Tool	Purpose	Install
Python 3.11+	Agent code	python.org
Node.js 18+	Frontend, n8n	nodejs.org
Docker	n8n self-hosted	docker.com
Git	Version control	git-scm.com
VS Code	Code editor	code.visualstudio.com
Claude Code	AI pair programmer	<code>npm install -g @anthropic-ai/claude-code</code>
Obsidian	Read the vault	obsidian.md
Postman	Test APIs	postman.com

Skill Progression Map

Day 1-2: Can EXPLAIN what agents are and how they work

Day 3: Can COMPARE frameworks and choose the right one

Day 4-5: Can BUILD a multi-agent system with RAG

Day 6: Can BUILD agent workflows in n8n

Day 7-8: Can DEMO and PITCH the project confidently

Day 9-10: Can ENHANCE and EXTEND the system for LaunchpadX

Key Knowledge Pages (Reading List)

Must-Read (Everyone)

1. Agentic AI - Master Guide
2. Types of AI Agents
3. Agent Design Patterns
4. Our AI Software Company
5. How to Build AI Agents

Should-Read (At least 3)

6. Multi-Agent Architectures

- 7. Agent Frameworks Comparison
- 8. Agent Protocols - MCP and A2A
- 9. RAG - Retrieval Augmented Generation
- 10. AI Agent Market - Competitors

Nice-to-Know

- 11. Voice Agents
- 12. Call Agents
- 13. Agentic AI Use Cases

[See Project Vision](#) | [Roadmap](#) | [Our AI Software Company](#)

**schedule #training #hackathon
#launchpadx #SIH**
